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Attention Is All

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摘要

主导的序列转导模型基于复杂的循环或
器和解码器。表现最佳的模型还通过注
们提出了一种新的简单网络架构——Tr
制，摒弃了循环和卷积。在两个机器翻
质量上更优越，同时更易于并行化，且
WMT 2014英语-德语翻译任务上达到
结果（包括集成模型）提高了超过2个
译任务上，我们的模型在八块GPU上训
SOTA BLEU分数41.8，其训练成本仅
们通过将其成功应用于英语短语结构分
他任务的泛化能力良好，无论是使用大

[*]平等贡献。列表顺序是随机的。Jakob 提议用自
Ashish 与 Illia 设计并实现了第一个 Transformer 模
Noam 提议了缩放点积注意力、多头注意力和无参数的
Niki 设计、实现、调整和评估了我们原始代码库和 ten
的模型变体，负责我们最初的代码库以及高效的推理和
设计和实现 tensor2tensor 的各个部分，替换了我们
们的研究。

[†]在 Google Brain 期间完成的工作。
[‡]在 Google Research 期间完成的工作。

第 31 届神经信息处理系统大会 (NIPS 2017)，长滩

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Abstr

The dominant sequence transduction mo
convolutional neural networks that includ
performing models also connect the enc
mechanism. We propose a new simple
based solely on attention mechanisms, disp
entirely. Experiments on two machine tr
be superior in quality while being more p
less time to train. Our model achieves 2
to-German translation task, improving o
ensembles, by over 2 BLEU. On the WMT
our model establishes a new single-model s
training for 3.5 days on eight GPUs, a sm
best models from the literature. We show t
other tasks by applying it successfully to
large and limited training data.

[*]Equal contribution. Listing order is random. Jakob pr
the effort to evaluate this idea. Ashish, with Illia, design
has been crucially involved in every aspect of this work. N
attention and the parameter-free position representation
detail. Niki designed, implemented, tuned and evaluated
tensor2tensor. Llion also experimented with novel model
efficient inference and visualizations. Lukasz and Aidan s
implementing tensor2tensor, replacing our earlier codebas
our research.

[†]Work performed while at Google Brain.
[‡]Work performed while at Google Research.

31st Conference on Neural Information Processing Syste

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ll You Need

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或卷积神经网络，这些网络包括编码
注意力机制连接编码器器和解码器。我
Transformer，它完全基于注意力机
翻译任务上的实验表明，这些模型在
且训练时间显著减少。我们的模型在
到了28.4的BLEU分数，比现有的最佳
个BLEU。在WMT 2014英语-法语翻
训练3.5天后，建立了一个新的单模型
仅为文献中最优模型的极小一部分。我
分析任务，证明了Transformer对其
大量还是有限训练数据。

自注意力机制替换 RNN，并开始评估这个想法。
模型，并在这项工作的各个方面都发挥了关键作用。
的位置表示，并成为参与几乎所有细节的另一位人物。
tensor2tensor 中的无数模型变体。Llion 也实验了新
和可视化。Lukasz 和 Aidan 花费了无数漫长的日子
们早期的代码库，极大地提高了结果并极大地加速了我

滩，CA，美国。

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All You Need

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stract

odels are based on complex recurrent or
lude an encoder and a decoder. The best
ncoder and decoder through an attention
le network architecture, the Transformer,
ispensing with recurrence and convolutions
e translation tasks show these models to
e parallelizable and requiring significantly
s 28.4 BLEU on the WMT 2014 English-
over the existing best results, including
T 2014 English-to-French translation task,
el state-of-the-art BLEU score of 41.8 after
small fraction of the training costs of the
w that the Transformer generalizes well to
to English constituency parsing both with

b proposed replacing RNNs with self-attention and started
igned and implemented the first Transformer models and
. Noam proposed scaled dot-product attention, multi-head
on and became the other person involved in nearly every
ted countless model variants in our original codebase and
del variants, was responsible for our initial codebase, and
n spent countless long days designing various parts of and
base, greatly improving results and massively accelerating

ystems (NIPS 2017), Long Beach, CA, USA.

1 简介

循环神经网络，特别是长短期记忆 [13] 和门控
模和语言建模、机器翻译 [35, 2, 5]等转换问题
力以拓展循环语言模型和编码器-解码器架构 [3

循环模型通常沿着输入和输出序列的符号位置
生成一个隐藏状态序列 h_t，作为先前隐藏状态
序性质在训练示例内禁止并行化，这在较长的
制了跨示例的批处理。最近的工作通过分解技
著改进，同时后者还提高了模型性能。然而，

注意力机制已成为各种任务中引人入胜的序列
考虑输入或输出序列中它们距离的情况下建模
类注意力机制与循环网络一起使用。

在这项工作中，我们提出了 Transformer，一
输入和输出之间全局依赖关系的模型架构。Tra
经过在八块 P100 GPU 上训练仅十二小时后，

2 背景

减少顺序计算的目标也构成了Extended Neural
础，所有这些模型都使用卷积神经网络作为基本
表示。在这些模型中，将两个任意输入或输出位
间的距离而增加，对于ConvS2S是线性增加，对
之间的依赖关系变得更加困难 [12]。在Transfor
致了由于对注意力加权位置进行平均而导致的有
注意力机制来抵消这种效应。

自注意力机制，有时称为序列内注意力机制，
以便计算序列的表示。自注意力机制已成功应
文本蕴涵和学习任务无关的句子表示 [4, 27, 28,

端到端记忆网络基于循环注意力机制，而不是
问答和语言建模任务上表现良好 [34]。

据我们所知，然而，Transformer 是第一个完
转换模型，而不使用序列对齐的 RNN 或卷积
Transformer，论证自注意力及其相对于 [17,

3 模型架构

大多数具有竞争力的神经序列转换模型都具有
器将符号表示的输入序列 $(x_1, ..., x_n)$ 映射到
解码器随后逐个元素地生成符号输出序列 $(y_1,$
的，在生成下一个符号时将先前生成的符号作

1 Introduction

Recurrent neural networks, long short-term memo
in particular, have been firmly established as state
transduction problems such as language modeling
efforts have since continued to push the boundaries o
architectures [38, 24, 15].

Recurrent models typically factor computation alo
sequences. Aligning the positions to steps in comp
states h_t, as a function of the previous hidden state
sequential nature precludes parallelization within tra
sequence lengths, as memory constraints limit batc
significant improvements in computational efficienc
computation [32], while also improving model per
constraint of sequential computation, however, rem

Attention mechanisms have become an integral part
tion models in various tasks, allowing modeling of
the input or output sequences [2, 19]. In all but a fe
are used in conjunction with a recurrent network.

In this work we propose the Transformer, a model
relying entirely on an attention mechanism to draw
The Transformer allows for significantly more para
translation quality after being trained for as little as

2 Background

The goal of reducing sequential computation also fo
[16], ByteNet [18] and ConvS2S [9], all of which us
block, computing hidden representations in parallel f
the number of operations required to relate signals fr
in the distance between positions, linearly for Conv
it more difficult to learn dependencies between di
reduced to a constant number of operations, albeit
to averaging attention-weighted positions, an effe
described in section 3.2.

Self-attention, sometimes called intra-attention is an
of a single sequence in order to compute a represe
used successfully in a variety of tasks including rea
textual entailment and learning task-independent se

End-to-end memory networks are based on a recu
aligned recurrence and have been shown to perform
language modeling tasks [34].

To the best of our knowledge, however, the Tran
entirely on self-attention to compute representation
aligned RNNs or convolution. In the following sect
self-attention and discuss its advantages over mode

3 Model Architecture

Most competitive neural sequence transduction mo
Here, the encoder maps an input sequence of sy
of continuous representations $\mathbf{z} = (z_1, ..., z_n)$.
sequence $(y_1, ..., y_m)$ of symbols one element at a
[10], consuming the previously generated symbols

控循环 [7] 神经网络，已被牢固确立为序列建
题的最先进方法。此后，人们持续投入大量努
[38, 24, 15]的边界。

置进行计算。将位置与计算时间步对齐，它们
态 h_{t-1} 和位置 t 输入的函数。这种固有的顺
的序列长度时变得至关重要，因为内存限制限
技巧 [21]和条件计算 [32]，实现了计算效率的显
，顺序计算的约束仍然存在。

列建模和转换模型的重要组成部分，允许在不
模依赖关系 [2, 19]。但在少数情况下 [27], ，此

一种避免递归并完全依赖于注意力机制来绘制
ransformer 允许显著更多的并行化，并且在
，可以达到翻译质量的新状态。

al GPU[16], ByteNet [18] 和ConvS2S [9], 的基
本构建块，并行计算所有输入和输出位置的隐藏
位置的信号关联起来所需的操作次数随着位置之
对于ByteNet是对数增加。这使得学习远处位置
ormer中，这减少到常数次数的操作，尽管这导
有效分辨率降低，我们通过第3.2节中描述的多头

，是一种关联单个序列不同位置的注意力机制，
应用于多种任务，包括阅读理解、抽象摘要、
28, 22]。

是序列对齐的递归，并且已被证明在简单语言

完全依赖自注意力来计算其输入和输出表示的
。在以下几节中，我们将描述
17, 18] 和 [9] 等模型的优点。

有编码器-解码器结构 [5, 2, 35]。在这里，编码
到连续表示的序列 $\mathbf{z} = (z_1, ..., z_n)$。给定 $\mathbf{z}$,
$_1$, ..., y_m)。在每一步，模型都是自回归[10],
作为附加输入进行消耗。

mory [13] and gated recurrent [7] neural networks
te of the art approaches in sequence modeling and
ing and machine translation [35, 2, 5]. Numerous
s of recurrent language models and encoder-decoder

along the symbol positions of the input and output
mputation time, they generate a sequence of hidden
te h_{t-1} and the input for position t. This inherently
training examples, which becomes critical at longer
atching across examples. Recent work has achieved
ncy through factorization tricks [21] and conditional
performance in case of the latter. The fundamental
emains.

part of compelling sequence modeling and transduc-
of dependencies without regard to their distance in
few cases [27], however, such attention mechanisms
.

del architecture eschewing recurrence and instead
raw global dependencies between input and output.
arallelization and can reach a new state of the art in
as twelve hours on eight P100 GPUs.

o forms the foundation of the Extended Neural GPU
use convolutional neural networks as basic building
lel for all input and output positions. In these models,
ls from two arbitrary input or output positions grows
nvS2S and logarithmically for ByteNet. This makes
distant positions [12]. In the Transformer this is
beit at the cost of reduced effective resolution due
ffect we counteract with Multi-Head Attention as

s an attention mechanism relating different positions
esentation of the sequence. Self-attention has been
reading comprehension, abstractive summarization,
t sentence representations [4, 27, 28, 22].

ecurrent attention mechanism instead of sequence-
rm well on simple-language question answering and

ransformer is the first transduction model relying
ions of its input and output without using sequence-
ections, we will describe the Transformer, motivate
dels such as [17, 18] and [9].

models have an encoder-decoder structure [5, 2, 35].
symbol representations $(x_1, ..., x_n)$ to a sequence
. Given $\mathbf{z}$, the decoder then generates an output
t a time. At each step the model is auto-regressive
ols as additional input when generating the next.

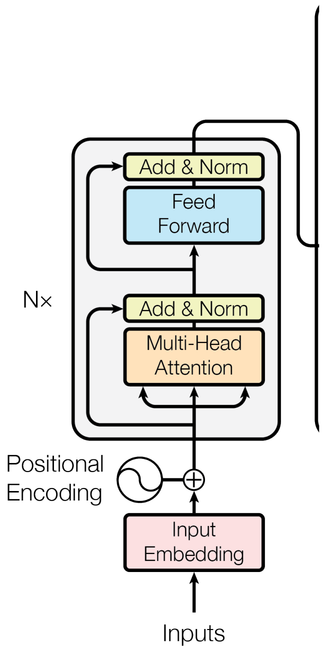


图1：Transformer -

Transformer 使用堆叠的自注意力和逐点、全
和解码器，如图1的左半部和右半部所示。

3.1 EncoderandDecoder Stacks

编码器： 编码器由一个堆叠的 $N = 6$ 相同层
注意力机制，第二个是一个简单、位置相关的
残差连接 [11]，然后进行层归一化 [1]。也就
$\mathrm{Sublayer}(x))$，其中 $\mathrm{Sublayer}(x)$ 是子层本身
中的所有子层以及嵌入层都产生维度为 $d_{\text{model}} =$

解码器： 解码器也由一个堆叠的 $N = 6$ 相同
解码器还插入一个第三个子层，该子层对编码
似，我们在每个子层周围使用残差连接，然后
的自注意力子层，以防止位置关注后续位置。
实，确保位置 i 的预测只能依赖于位置小于 i

3.2 注意力

一个注意力函数可以被描述为将一个查询和一组键值
是向量。输出计算为值的加权求和

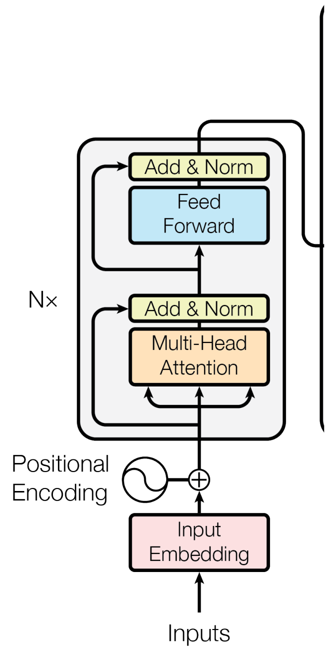


Figure 1: The Transforme

The Transformer follows this overall architecture u
connected layers for both the encoder and decoder
respectively.

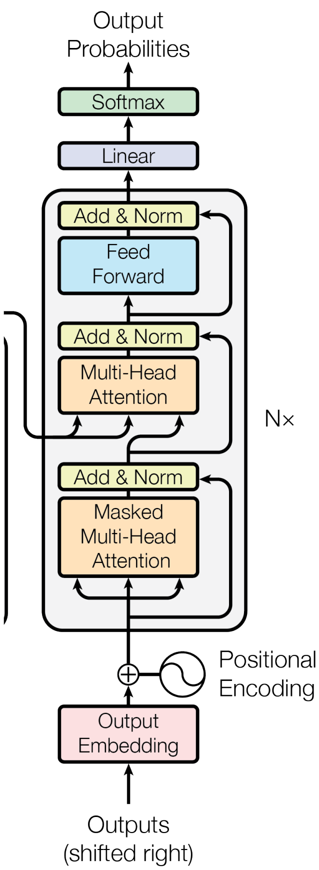
3.1 Encoder and Decoder Stacks

Encoder: The encoder is composed of a stack
sub-layers. The first is a multi-head self-attention
wise fully connected feed-forward network. We e
the two sub-layers, followed by layer normalizati
$\mathrm{LayerNorm}(x + \mathrm{Sublayer}(x))$, where $\mathrm{Sublayer}(x$
itself. To facilitate these residual connections, all su
layers, produce outputs of dimension $d_{\text{model}} = 512$.

Decoder: The decoder is also composed of a stack
sub-layers in each encoder layer, the decoder inser
attention over the output of the encoder stack. Simila
around each of the sub-layers, followed by layer n
sub-layer in the decoder stack to prevent position
masking, combined with fact that the output embed
predictions for position i can depend only on the kn

3.2 Attention

An attention function can be described as mapping
where the query, keys, values, and output are all vec



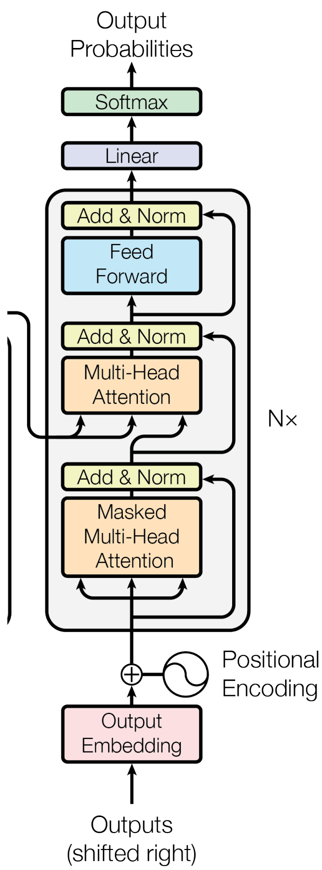
- 模型架构。

全连接层遵循这种整体架构，分别用于编码器

层组成。每一层有两个子层。第一个是多头自
的全连接前馈网络。我们在每个子层周围使用
就是说，每个子层的输出是 LayerNorm(x +
身实现的函数。为了方便这些残差连接，模型
$_{1} = 512$的输出。

同层组成。除了每个编码器层中的两个子层外，
码器堆栈的输出执行多头注意力。与编码器类
后进行层归一化。我们还修改了解码器堆栈中
。这种掩码，结合输出嵌入偏移一个位置的事
i的已知输出。

值对映射到一个输出，其中查询、键、值和输出都



rmer - model architecture.

re using stacked self-attention and point-wise, fully
der, shown in the left and right halves of Figure 1,

k of $N = 6$ identical layers. Each layer has two
n mechanism, and the second is a simple, position-
employ a residual connection [11] around each of
ation [1]. That is, the output of each sub-layer is
r(x) is the function implemented by the sub-layer
l sub-layers in the model, as well as the embedding
12.

ack of $N = 6$ identical layers. In addition to the two
serts a third sub-layer, which performs multi-head
ilar to the encoder, we employ residual connections
r normalization. We also modify the self-attention
ions from attending to subsequent positions. This
beddings are offset by one position, ensures that the
known outputs at positions less than i.

ng a query and a set of key-value pairs to an output,
vectors. The output is computed as a weighted sum

3

Scaled Dot-Product Attention

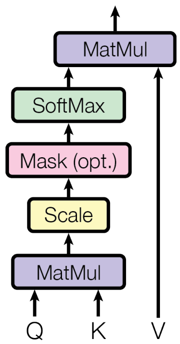


图2：(左) Scaled Dot-Product Attention。(右) 层组成。

其中分配给每个值的权重由查询与相应键的兼

3.2.1 缩放点积注意力

我们将这种注意力称为 "Scaled Dot-Product 和键，以及维度为 d_v 的值组成。我们将查询与 用 softmax 函数以获得权重。

在实践中，我们同时在一组查询上计算注意力 被打包成矩阵 K 和 V。输出的矩阵计算如下

$$\mathrm{Attention}(Q, K, V) =$$

两种最常用的注意力函数是加性注意力 [2], 和 法相同，除了缩放因子 $\frac{1}{\sqrt{d_k}}$。加性注意力使用一 虽然两者在理论复杂度上相似，但在实践中， 可以使用高度优化的矩阵乘法代码来实现。

虽然对于较小的 d_k 值，这两种机制表现相似， 的情况下优于点积注意力。我们推测，对于较 softmax函数推入梯度极小的区域 [4]。为了抵消

3.2.2 Multi-Head Attention

与其使用 d_{model}-维的键、值和查询执行单个注 过不同的、学习到的线性投影线性投影 h 次到 些投影版本的查询、键和值上并行执行注意力

[4]为了说明为什么点积会变得很大，假设 q 和 k 它们的点积 $q \cdot k = \sum_{i=1}^{d_k} q_i k_i$的均值为0，方差为 d_k

Scaled Dot-Product Attention

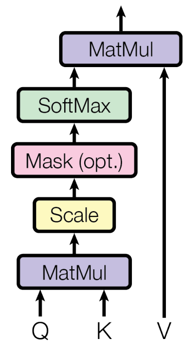


Figure 2: (left) Scaled Dot-Product Attention. (r attention layers running in parallel.

of the values, where the weight assigned to each val query with the corresponding key.

3.2.1 Scaled Dot-Product Attention

We call our particular attention "Scaled Dot-Prod queries and keys of dimension d_k, and values of di query with all keys, divide each by $\sqrt{d_k}$, and apply values.

In practice, we compute the attention function on into a matrix Q. The keys and values are also pack the matrix of outputs as:

$$\mathrm{Attention}(Q, K, V) =$$

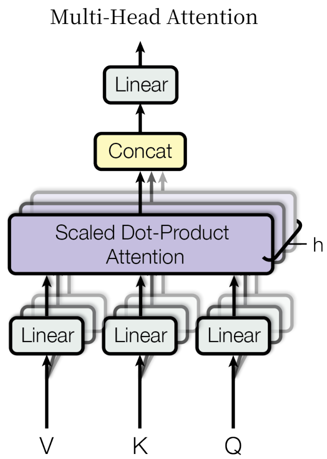
The two most commonly used attention functions a plicative) attention. Dot-product attention is identic of $\frac{1}{\sqrt{d_k}}$. Additive attention computes the compatibi a single hidden layer. While the two are similar in much faster and more space-efficient in practice, sin matrix multiplication code.

While for small values of d_k the two mechanisms p dot product attention without scaling for larger val d_k, the dot products grow large in magnitude, pushi extremely small gradients [4]. To counteract this effe

3.2.2 Multi-Head Attention

Instead of performing a single attention function w we found it beneficial to linearly project the queries linear projections to d_k, d_k and d_v dimensions, res queries, keys and values we then perform the attenti

[4]To illustrate why the dot products get large, assume th variables with mean 0 and variance 1. Then their dot pro



) Multi-Head Attention 由多个并行运行的注意力

兼容性函数计算得出。

ct Attention" (图2)。输入由维度为 d_k 的查询
与所有键的点积相乘，除以 $\sqrt{d_k}$，并对值应

力函数，将它们打包成一个矩阵 Q。键和值也
下:

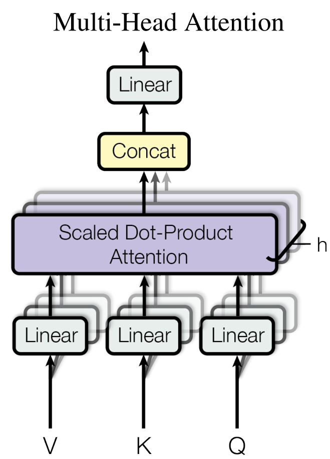
$$) = \mathrm{softmax}(\frac{QK^T}{\sqrt{d_k}})V \tag{1}$$

点积（乘性）注意力。点积注意力与我们的算
一个单隐藏层的前馈网络来计算兼容性函数。
，点积注意力要快得多，也更节省空间，因为它

，但对于较大的 d_k [3]值，加性注意力在无缩放
大的d_k值，点积的幅度会变得很大，将
消这种效应，我们通过 $\frac{1}{\sqrt{d_k}}$对点积进行缩放。

意力函数，我们发现将查询、键和值分别通
到 d_k、 d_k 和 d_v 维是有益的。然后，我们在这
力函数，产生 d_v-维

k 的分量是均值为0、方差为1的独立随机变量。那么
d_k。



(right) Multi-Head Attention consists of several

value is computed by a compatibility function of the

oduct Attention" (Figure 2). The input consists of
dimension d_v. We compute the dot products of the
ply a softmax function to obtain the weights on the

n a set of queries simultaneously, packed together
cked together into matrices K and V. We compute

$$) = \mathrm{softmax}(\frac{QK^T}{\sqrt{d_k}})V \tag{1}$$

s are additive attention [2], and dot-product (multi-
ntical to our algorithm, except for the scaling factor
tibility function using a feed-forward network with
in theoretical complexity, dot-product attention is
since it can be implemented using highly optimized

s perform similarly, additive attention outperforms
values of d_k [3]. We suspect that for large values of
shing the softmax function into regions where it has
ffect, we scale the dot products by $\frac{1}{\sqrt{d_k}}$.

n with d_{model}-dimensional keys, values and queries,
ries, keys and values h times with different, learned
respectively. On each of these projected versions of
ention function in parallel, yielding d_v-dimensional

e that the components of q and k are independent random
product, $q \cdot k = \sum_{i=1}^{d_k} q_i k_i$, has mean 0 and variance d_k.

输出值。这些值被连接起来，再次投影，从而

多头注意力允许模型在多个位置同时关注来自不同表
作会抑制这种能力。

$$\mathrm{MultiHead}(Q, K, V) = \mathrm{Conc}$$
$$\text{where } \mathrm{head_i} = \mathrm{Atten}$$

其中投影是参数矩阵 $W_i^Q \in \mathbb{R}^{d_{\text{model}} \times d_k}$, $W_i^K \in \mathbb{R}^{d_{\text{mo}}}$

在这项工作中，我们采用 $h = 8$ 个并行注意
$d_k = d_v = d_{\text{model}}/h = 64$。由于每个头的维度降

3.2.3 注意力在我们模型中的应用

Transformer 以三种不同的方式使用多头注意

- 在 "编码器-解码器注意力" 层中，查询
码器的输出。这允许解码器中的每个位
到序列模型（如 [38, 2, 9]）中的典型编码

- 编码器包含自注意力层。在自注意力层
在这种情况下，编码器中前一层的输出
一层的所有位置。

- 类似地，解码器中的自注意力层允许解
包括该位置本身。我们需要防止解码器
在缩放点积注意力内部通过屏蔽（设置
有值来实现这一点。见图 2。

3.3 Position-wise Feed-Forward Networks

除了注意力子层之外，我们编码器和解码器中
网络分别且相同地应用于每个位置。这由两个

$$\mathrm{FFN}(x) = \max(0, x$$

虽然线性变换在不同位置上是相同的，但它们
方式是两个卷积，卷积核大小为1。输入和输出
$d_{ff} = 2048$。

3.4 嵌入和Softmax

与其他序列转导模型类似，我们使用学习到的
d_{model}的向量。我们还使用通常的学习到的线性
测的下一个标记概率。在我们的模型中，我们
享相同的权重矩阵，类似于 [30]。在嵌入层中

output values. These are concatenated and once a
depicted in Figure 2.

Multi-head attention allows the model to jointly att
subspaces at different positions. With a single atten

$$\mathrm{MultiHead}(Q, K, V) = \mathrm{Conc}$$
$$\text{where } \mathrm{head_i} = \mathrm{Atten}$$

Where the projections are parameter matrices $W_i^Q \in$
and $W^O \in \mathbb{R}^{hd_v \times d_{\text{model}}}$.

In this work we employ $h = 8$ parallel attentio
$d_k = d_v = d_{\text{model}}/h = 64$. Due to the reduced dim
is similar to that of single-head attention with full d

3.2.3 Applications of Attention in our Model

The Transformer uses multi-head attention in three

- In "encoder-decoder attention" layers, the
and the memory keys and values come fr
position in the decoder to attend over all p
typical encoder-decoder attention mecha
[38, 2, 9].
- The encoder contains self-attention layers
and queries come from the same place, in
encoder. Each position in the encoder can
encoder.
- Similarly, self-attention layers in the decod
all positions in the decoder up to and incl
information flow in the decoder to preserve
inside of scaled dot-product attention by m
of the softmax which correspond to illegal

3.3 Position-wise Feed-Forward Networks

In addition to attention sub-layers, each of the lay
connected feed-forward network, which is applied
consists of two linear transformations with a ReLU

$$\mathrm{FFN}(x) = \max(0, x$$

While the linear transformations are the same across
from layer to layer. Another way of describing
The dimensionality of input and output is d_{model}
$d_{ff} = 2048$.

3.4 Embeddings and Softmax

Similarly to other sequence transduction models,
tokens and output tokens to vectors of dimension d_{m}
mation and softmax function to convert the decode
our model, we share the same weight matrix betwee
linear transformation, similar to [30]. In the embedd

而得到最终值，如图2所示。

表示子空间的信息。使用单个注意力头时，平均操

$\mathrm{ncat}(\mathrm{head_1}, ..., \mathrm{head_h})W^O$
$\mathrm{tention}(QW_i^Q, KW_i^K, VW_i^V)$

${}^{d_{\text{model}} \times d_k}$, $W_i^V \in \mathbb{R}^{d_{\text{model}} \times d_v}$ 和 $W^O \in \mathbb{R}^{hd_v \times d_{\text{model}}}$。

意力层，或头。对于每一个，我们使用
降低，总计算成本与全维度的单头注意力相似。

意力:

询来自前一个解码器层，而记忆键和值来自编
置关注输入序列中的所有位置。这模拟了序列
码器-解码器注意力机制。

层中，所有的键、值和查询都来自同一个地方，
。编码器中的每个位置都可以关注编码器前

解码器中的每个位置关注解码器中所有位置，
中的左向信息流动以保持自回归特性。我们
为 $-\infty$）softmax 输入中对应非法连接的所

中的每一层都包含一个全连接的前馈网络，该
个线性变换和一个 ReLU 激活函数组成。

$$, xW_1 + b_1)W_2 + b_2 \quad (2)$$

在层与层之间使用不同的参数。另一种描述
出的维度是 $d_{\text{model}} = 512$，内层的维度是

的嵌入将输入标记和输出标记转换为维度为
性转换和softmax函数将解码器输出转换为预
们在两个嵌入层和预softmax线性转换之间共
，我们将这些权重乘以 $\sqrt{d_{\text{model}}}$。

e again projected, resulting in the final values, as

attend to information from different representation
ttention head, averaging inhibits this.

$\mathrm{ncat}(\mathrm{head_1}, ..., \mathrm{head_h})W^O$
$\mathrm{tention}(QW_i^Q, KW_i^K, VW_i^V)$

$\in \mathbb{R}^{d_{\text{model}} \times d_k}$, $W_i^K \in \mathbb{R}^{d_{\text{model}} \times d_k}$, $W_i^V \in \mathbb{R}^{d_{\text{model}} \times d_v}$

tion layers, or heads. For each of these we use
imension of each head, the total computational cost
ll dimensionality.

l

ree different ways:

the queries come from the previous decoder layer,
from the output of the encoder. This allows every
ll positions in the input sequence. This mimics the
hanisms in sequence-to-sequence models such as

ers. In a self-attention layer all of the keys, values
, in this case, the output of the previous layer in the
an attend to all positions in the previous layer of the

coder allow each position in the decoder to attend to
cluding that position. We need to prevent leftward
rve the auto-regressive property. We implement this
masking out (setting to $-\infty$) all values in the input
gal connections. See Figure 2.

layers in our encoder and decoder contains a fully
ed to each position separately and identically. This
LU activation in between.

$$, xW_1 + b_1)W_2 + b_2 \quad (2)$$

oss different positions, they use different parameters
ng this is as two convolutions with kernel size 1.
${}_{\text{del}} = 512$, and the inner-layer has dimensionality

ls, we use learned embeddings to convert the input
d_{model}. We also use the usual learned linear transfor-
der output to predicted next-token probabilities. In
ween the two embedding layers and the pre-softmax
edding layers, we multiply those weights by $\sqrt{d_{\text{model}}}$.

表1：不同层类型的最大路径长度、每层复杂度
示维度， k 是卷积的核大小， r 是受限自注意

层类型	每层复杂度
自注意力	$O(n^2 \cdot d)$
循环	$O(n \cdot d^2)$
卷积	$O(k \cdot n \cdot d^2)$
自注意力（受限）	$O(r \cdot n \cdot d)$

3.5 位置编码

由于我们的模型不包含递归和卷积，为了使模
关于序列中标记的相对或绝对位置的信息。为
嵌入中添加了“位置编码”。位置编码的维度
有很多选择，包括学习和固定的 [9]。

在这项工作中，我们使用不同频率的正弦和余

$$PE_{(pos,2i)} = sin($$
$$PE_{(pos,2i+1)} = cos($$

where pos 是位置， i 是维度。也就是说，位
2π 到 $10000 \cdot 2\pi$ 形成一个几何级数。我们选择
地学习通过相对位置来关注，因为对于任何固
的线性函数。
我们还尝试使用学习的位置嵌入 [9]，发现两
我们选择正弦版本是因为它可能允许模型推广

4 为什么是自注意力

在本节中，我们将自注意力层与通常用于将一
到另一个等长序列 $(z_1,$ ... $z_n)$ 的层进行比较，
为了说明我们使用自注意力的原因，我们考虑

一个是每层的总计算复杂度。另一个是可以并行

第三个是网络中长期依赖之间的路径长度。学
键挑战。影响学习这种依赖能力的一个关键因
径长度。输入和输出序列中任何位置组合之间
因此，我们还比较了由不同层类型组成的网络
长度。

如表1所示，自注意力层通过固定数量的顺序执
个顺序操作。在计算复杂度方面，当序列

Table 1: Maximum path lengths, per-layer complexi
for different layer types. n is the sequence length,
size of convolutions and r the size of the neighborh

Layer Type	Complexity per L
Self-Attention	$O(n^2 \cdot d)$
Recurrent	$O(n \cdot d^2)$
Convolutional	$O(k \cdot n \cdot d^2)$
Self-Attention (restricted)	$O(r \cdot n \cdot d)$

3.5 Positional Encoding

Since our model contains no recurrence and no conv
order of the sequence, we must inject some informa
tokens in the sequence. To this end, we add "positi
bottoms of the encoder and decoder stacks. The posi
as the embeddings, so that the two can be summed.
learned and fixed [9].

In this work, we use sine and cosine functions of di

$$PE_{(pos,2i)} = sin($$
$$PE_{(pos,2i+1)} = cos($$

where pos is the position and i is the dimension. Th
corresponds to a sinusoid. The wavelengths form a g
chose this function because we hypothesized it wo
relative positions, since for any fixed offset k, PE
PE_{pos}.

We also experimented with using learned positional
versions produced nearly identical results (see Ta
because it may allow the model to extrapolate to se
during training.

4 Why Self-Attention

In this section we compare various aspects of sel
tional layers commonly used for mapping one vari
$(x_1, ..., x_n)$ to another sequence of equal length (
layer in a typical sequence transduction encoder or
consider three desiderata.

One is the total computational complexity per layer
be parallelized, as measured by the minimum numb

The third is the path length between long-range de
dependencies is a key challenge in many sequence
ability to learn such dependencies is the length of t
traverse in the network. The shorter these paths bet
and output sequences, the easier it is to learn long-r
the maximum path length between any two input a
different layer types.

As noted in Table 1, a self-attention layer connects al
executed operations, whereas a recurrent layer re
computational complexity, self-attention layers are

度和最小顺序操作数。 n 是序列长度， d 是表
力中的邻域大小。

	顺序操作	最大路径长度
)	$O(1)$	$O(1)$
	$O(n)$	$O(n)$
^{2})	$O(1)$	$O(log_k(n))$
)	$O(1)$	$O(n/r)$

模型能够利用序列的顺序，我们必须注入一些
为此，我们在编码器和解码器堆栈底部的输入
度与嵌入相同，以便两者可以相加。位置编码

余弦函数：

$$n(pos/10000^{2i/d_{\text{model}}})$$
$$s(pos/10000^{2i/d_{\text{model}}})$$

位置编码的每个维度对应一个正弦波。波长从
择这个函数是因为我们假设它可以让模型容易
固定的偏移量 k， PE_{pos+k} 可以表示为 PE_{pos}

个版本产生的结果几乎相同（见表 3 行 (E)）。
到比训练中遇到的更长的序列长度。

一个变量长度的符号表示序列 $(x_1$， ... $x_n)$ 映射
例如典型序列转换编码器或解码器中的隐藏层。
虑了三个期望。

行化的计算量，以所需的最少顺序操作数来衡量。

学习长期依赖是许多序列转换任务中的一个关
因素是前向和后向信号在网络中必须穿越的路
间的路径越短，学习长期依赖就越容易 [12]。
络中任何两个输入和输出位置之间的最大路径

执行操作连接所有位置，而循环层需要 $O(n)$

lexity and minimum number of sequential operations
h, d is the representation dimension, k is the kernel
orhood in restricted self-attention.

r Layer	Sequential Operations	Maximum Path Length
)	$O(1)$	$O(1)$
)	$O(n)$	$O(n)$
^{2})	$O(1)$	$O(log_k(n))$
d)	$O(1)$	$O(n/r)$

onvolution, in order for the model to make use of the
mation about the relative or absolute position of the
sitional encodings" to the input embeddings at the
ositional encodings have the same dimension d_{model}
ed. There are many choices of positional encodings,

f different frequencies:

$$n(pos/10000^{2i/d_{\text{model}}})$$
$$s(pos/10000^{2i/d_{\text{model}}})$$

. That is, each dimension of the positional encoding
a geometric progression from 2π to $10000 \cdot 2\pi$. We
would allow the model to easily learn to attend by
E_{pos+k} can be represented as a linear function of

nal embeddings [9] instead, and found that the two
Table 3 row (E)). We chose the sinusoidal version
sequence lengths longer than the ones encountered

self-attention layers to the recurrent and convolu-
ariable-length sequence of symbol representations
h $(z_1, ..., z_n)$, with $x_i, z_i \in \mathbb{R}^d$, such as a hidden
or decoder. Motivating our use of self-attention we

yer. Another is the amount of computation that can
mber of sequential operations required.

dependencies in the network. Learning long-range
ce transduction tasks. One key factor affecting the
of the paths forward and backward signals have to
between any combination of positions in the input
g-range dependencies [12]. Hence we also compare
t and output positions in networks composed of the

s all positions with a constant number of sequentially
requires $O(n)$ sequential operations. In terms of
are faster than recurrent layers when the sequence

为了提高涉及非常长序列的任务的计算性能，
相应输出位置的大小为 r 的邻域。这将使最大
的工作中进一步研究这种方法。

一个卷积层，其核宽度为 $k < n$，并不连接所
卷积层堆叠，在连续核的情况下，或者 $O(log_k($
网络中任意两个位置之间最长路径的长度。卷
然而，可分离卷积 [6],，大大降低了复杂度，
分离卷积的复杂度也等于自注意力层和逐点前

作为附加好处，自注意力机制可以产生更可解
并在附录中展示和讨论了示例。不仅单个注意
力头似乎表现出与句子句法和语义结构相关的

5 训练

本节描述了我们模型的训练机制。

5.1 训练数据和批处理

我们在标准的WMT 2014英德数据集上进行了
用字节对编码 [3], 进行编码，具有约37000个共
了显著更大的WMT 2014英法语数据集，包含3
的词汇表 [38]。句子对按近似序列长度进行批
源标记和25000个目标标记的句子对。

5.2 硬件和计划

我们在一台机器上的8个NVIDIA P100 GPU上
述的超参数的基础模型，每个训练步骤大约需
100,000步或12小时。对于我们的大型模型（如
型训练了300,000步（3.5天）。

5.3 优化器

我们使用了 Adam 优化器 [20]，并使用 $\beta_1 = 0.9$、
学习率，根据公式：

$$lrate = d_{\text{model}}^{-0.5} \cdot \min(step\_num^{-0.5},$$

这对应于在前 $warmup\_steps$ 个训练步骤中线
成比例减小。我们使用了 $warmup\_steps = 400$

5.4 正则化

我们在训练过程中采用三种类型的正则化：

length n is smaller than the representation dimen
sentence representations used by state-of-the-art mo
[38] and byte-pair [31] representations. To improve
very long sequences, self-attention could be restricte
the input sequence centered around the respective ou
path length to $O(n/r)$. We plan to investigate this a

A single convolutional layer with kernel width $k <$
positions. Doing so requires a stack of $O(n/k)$ conv
or $O(log_k(n))$ in the case of dilated convolutions
between any two positions in the network. Convol
recurrent layers, by a factor of k. Separable conv
considerably, to $O(k \cdot n \cdot d + n \cdot d^2)$. Even with
convolution is equal to the combination of a self-att
the approach we take in our model.

As side benefit, self-attention could yield more interp
from our models and present and discuss examples i
heads clearly learn to perform different tasks, many
and semantic structure of the sentences.

5 Training

This section describes the training regime for our m

5.1 Training Data and Batching

We trained on the standard WMT 2014 English-
sentence pairs. Sentences were encoded using byt
target vocabulary of about 37000 tokens. For Englis
2014 English-French dataset consisting of 36M sen
vocabulary [38]. Sentence pairs were batched togeth
batch contained a set of sentence pairs containing
target tokens.

5.2 Hardware and Schedule

We trained our models on one machine with 8 N
the hyperparameters described throughout the pape
trained the base models for a total of 100,000 steps
bottom line of table 3), step time was 1.0 seconds.
(3.5 days).

5.3 Optimizer

We used the Adam optimizer [20] with $\beta_1 = 0.9$, β
rate over the course of training, according to the for

$$lrate = d_{\text{model}}^{-0.5} \cdot \min(step\_num^{-0.5},$$

This corresponds to increasing the learning rate lin
and decreasing it thereafter proportionally to the i
$warmup\_steps = 4000$.

5.4 Regularization

We employ three types of regularization during trai

，自注意力可以限制为仅考虑输入序列中围绕
大路径长度增加到 $O(n/r)$。我们计划在未来

ensionality d, which is most often the case with
models in machine translations, such as word-piece
ove computational performance for tasks involving
icted to considering only a neighborhood of size r in
e output position. This would increase the maximum
is approach further in future work.

所有输入和输出位置。这样做需要 $O(n/k)$ 个
$g_k(n))$ 个，在扩张卷积的情况下 [18]，，增加
卷积层通常比循环层更昂贵，成本高出 k 倍。
，至 $O(k \cdot n \cdot d + n \cdot d^2)$。即使有 $k = n$，可
前馈层的组合，这是我们模型采用的方法。

$< n$ does not connect all pairs of input and output
onvolutional layers in the case of contiguous kernels,
ns [18], increasing the length of the longest paths
olutional layers are generally more expensive than
nvolutions [6], however, decrease the complexity
ith $k = n$, however, the complexity of a separable
-attention layer and a point-wise feed-forward layer,

解释的模型。我们检查了模型中的注意力分布，
意力头清楚地学会了执行不同任务，许多注意
的行为。

erpretable models. We inspect attention distributions
les in the appendix. Not only do individual attention
ny appear to exhibit behavior related to the syntactic

r models.

了训练，该数据集包含约450万对句子。句子使
共享的源-目标词汇。对于英法语，我们使用
含36M个句子，并将标记分割成一个32000词
批处理。每个训练批次包含一组包含约25000个

h-German dataset consisting of about 4.5 million
byte-pair encoding [3], which has a shared source-
glish-French, we used the significantly larger WMT
sentences and split tokens into a 32000 word-piece
ether by approximate sequence length. Each training
ng approximately 25000 source tokens and 25000

上训练了我们的模型。对于我们使用全文中描
要0.4秒。我们训练基础模型总共进行了
如表3底部所述），步骤时间为1.0秒。大型模

NVIDIA P100 GPUs. For our base models using
aper, each training step took about 0.4 seconds. We
ps or 12 hours. For our big models,(described on the
ds. The big models were trained for 300,000 steps

9、 $\beta_2 = 0.98$ 和 $\epsilon = 10^{-9}$。我们在训练过程中调整

$${}^{0.5}, step\_num \cdot warmup\_steps^{-1.5}) \qquad (3)$$

线性增加学习率，之后按步骤数的平方根倒数
000。

9, $\beta_2 = 0.98$ and $\epsilon = 10^{-9}$. We varied the learning
formula:

$${}^{0.5}, step\_num \cdot warmup\_steps^{-1.5}) \qquad (3)$$

linearly for the first $warmup\_steps$ training steps,
e inverse square root of the step number. We used

training:

Table 2: The Transformer achieves better BLEU sc
English-to-German and English-to-French newstest

Model	EN-D
ByteNet [18]	23.7
Deep-Att + PosUnk [39]	
GNMT + RL [38]	24.6
ConvS2S [9]	25.1
MoE [32]	26.0
Deep-Att + PosUnk Ensemble [39]	
GNMT + RL Ensemble [38]	26.3
ConvS2S Ensemble [9]	26.3
Transformer (base model)	27.3
Transformer (big)	**28.4**

Residual Dropout We apply dropout [33] to the sub-layer input and normalized. In addition, we appl positional encodings in both the encoder and deco $P_{drop} = 0.1$.

Label Smoothing During training, we employe hurts perplexity, as the model learns to be more uns

6 Results

6.1 Machine Translation

On the WMT 2014 English-to-German translation ta in Table 2) outperforms the best previously reported BLEU, establishing a new state-of-the-art BLEU s listed in the bottom line of Table 3. Training took surpasses all previously published models and ense the competitive models.

On the WMT 2014 English-to-French translation tas outperforming all of the previously published singl previous state-of-the-art model. The Transformer dropout rate $P_{drop} = 0.1$, instead of 0.3.

For the base models, we used a single model obtai were written at 10-minute intervals. For the big m used beam search with a beam size of 4 and lengt were chosen after experimentation on the developme inference to input length + 50, but terminate early

Table 2 summarizes our results and compares our tra architectures from the literature. We estimate the n model by multiplying the training time, the number single-precision floating-point capacity of each GP

6.2 Model Variations

To evaluate the importance of different component in different ways, measuring the change in perfor

[5] We used values of 2.8, 3.7, 6.0 and 9.5 TFLOPS for

Table 2: The Transformer achieves better BLEU sc
English-to-German and English-to-French newstest

Model	EN-D
ByteNet [18]	23.7
Deep-Att + PosUnk [39]	
GNMT + RL [38]	24.6
ConvS2S [9]	25.1
MoE [32]	26.0
Deep-Att + PosUnk Ensemble [39]	
GNMT + RL Ensemble [38]	26.3
ConvS2S Ensemble [9]	26.3
Transformer (base model)	27.3
Transformer (big)	**28.4**

Residual Dropout We apply dropout [33] to the sub-layer input and normalized. In addition, we appl positional encodings in both the encoder and deco $P_{drop} = 0.1$.

Label Smoothing During training, we employe hurts perplexity, as the model learns to be more uns

6 Results

6.1 Machine Translation

On the WMT 2014 English-to-German translation ta in Table 2) outperforms the best previously reported BLEU, establishing a new state-of-the-art BLEU s listed in the bottom line of Table 3. Training took surpasses all previously published models and ense the competitive models.

On the WMT 2014 English-to-French translation tas outperforming all of the previously published singl previous state-of-the-art model. The Transformer dropout rate $P_{drop} = 0.1$, instead of 0.3.

For the base models, we used a single model obtai were written at 10-minute intervals. For the big m used beam search with a beam size of 4 and lengt were chosen after experimentation on the developme inference to input length + 50, but terminate early

Table 2 summarizes our results and compares our tra architectures from the literature. We estimate the n model by multiplying the training time, the number single-precision floating-point capacity of each GP

6.2 Model Variations

To evaluate the importance of different component in different ways, measuring the change in perfor

[5] We used values of 2.8, 3.7, 6.0 and 9.5 TFLOPS for

8

scores than previous state-of-the-art models on the test2014 tests at a fraction of the training cost.

BLEU		Training Cost (FLOPs)	
-DE	EN-FR	EN-DE	EN-FR
.75			
	39.2		$1.0 \cdot 10^{20}$
4.6	39.92	$2.3 \cdot 10^{19}$	$1.4 \cdot 10^{20}$
.16	40.46	$9.6 \cdot 10^{18}$	$1.5 \cdot 10^{20}$
.03	40.56	$2.0 \cdot 10^{19}$	$1.2 \cdot 10^{20}$
	40.4		$8.0 \cdot 10^{20}$
.30	41.16	$1.8 \cdot 10^{20}$	$1.1 \cdot 10^{21}$
.36	**41.29**	$7.7 \cdot 10^{19}$	$1.2 \cdot 10^{21}$
7.3	38.1	$\mathbf{3.3 \cdot 10^{18}}$	
8.4	**41.8**	$2.3 \cdot 10^{19}$	

he output of each sub-layer, before it is added to the pply dropout to the sums of the embeddings and the coder stacks. For the base model, we use a rate of

yed label smoothing of value $\epsilon_{ls} = 0.1$ [36]. This unsure, but improves accuracy and BLEU score.

n task, the big transformer model (Transformer (big) rted models (including ensembles) by more than 2.0 U score of 28.4. The configuration of this model is k 3.5 days on 8 P100 GPUs. Even our base model nsembles, at a fraction of the training cost of any of

task, our big model achieves a BLEU score of 41.0, gle models, at less than 1/4 the training cost of the er (big) model trained for English-to-French used

btained by averaging the last 5 checkpoints, which models, we averaged the last 20 checkpoints. We gth penalty $\alpha = 0.6$ [38]. These hyperparameters ment set. We set the maximum output length during ly when possible [38].

translation quality and training costs to other model e number of floating point operations used to train a ber of GPUs used, and an estimate of the sustained GPU [5].

ents of the Transformer, we varied our base model formance on English-to-German translation on the

for K80, K40, M40 and P100, respectively.

scores than previous state-of-the-art models on the test2014 tests at a fraction of the training cost.

BLEU		Training Cost (FLOPs)	
-DE	EN-FR	EN-DE	EN-FR
.75			
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Table 3: Variations on the Transformer architecture.
model. All metrics are on the English-to-German tr
perplexities are per-wordpiece, according to our byt
per-word perplexities.

	N	d_{model}	d_{ff}	h	d_k	d_v
base	6	512	2048	8	64	64
(A)				1	512	512
				4	128	128
				16	32	32
				32	16	16
(B)					16	
					32	
(C)	2					
	4					
	8					
		256			32	32
		1024			128	128
			1024			
			4096			
(D)						
(E)		positional embedding instead of s				
big	6	1024	4096	16		

development set, newstest2013. We used beam sea
checkpoint averaging. We present these results in T

In Table 3 rows (A), we vary the number of attention
keeping the amount of computation constant, as
attention is 0.9 BLEU worse than the best setting, q

In Table 3 rows (B), we observe that reducing the
suggests that determining compatibility is not ea
function than dot product may be beneficial. We furt
bigger models are better, and dropout is very helpful
sinusoidal positional encoding with learned positio
results to the base model.

6.3 English Constituency Parsing

To evaluate if the Transformer can generalize to ot
constituency parsing. This task presents specific cha
constraints and is significantly longer than the in
models have not been able to attain state-of-the-art

We trained a 4-layer transformer with $d_{model} = 102$
Penn Treebank [25], about 40K training sentences.
using the larger high-confidence and BerkleyParser
[37]. We used a vocabulary of 16K tokens for the
for the semi-supervised setting.

We performed only a small number of experiments
(section 5.4), learning rates and beam size on the S
remained unchanged from the English-to-German

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re. Unlisted values are identical to those of the base
n translation development set, newstest2013. Listed
byte-pair encoding, and should not be compared to

P_{drop}	ϵ_{ls}	train steps	PPL (dev)	BLEU (dev)	params $\times 10^6$
0.1	0.1	100K	4.92	25.8	65
			5.29	24.9	
			5.00	25.5	
			4.91	25.8	
			5.01	25.4	
			5.16	25.1	58
			5.01	25.4	60
			6.11	23.7	36
			5.19	25.3	50
			4.88	25.5	80
			5.75	24.5	28
			4.66	26.0	168
			5.12	25.4	53
			4.75	26.2	90
0.0			5.77	24.6	
0.2			4.95	25.5	
	0.0		4.67	25.3	
	0.2		5.47	25.7	
of sinusoids			4.92	25.7	
0.3		300K	**4.33**	**26.4**	213

search as described in the previous section, but no
n Table 3.

ion heads and the attention key and value dimensions,
as described in Section 3.2.2. While single-head
g, quality also drops off with too many heads.

the attention key size d_k hurts model quality. This
easy and that a more sophisticated compatibility
further observe in rows (C) and (D) that, as expected,
ful in avoiding over-fitting. In row (E) we replace our
tional embeddings [9], and observe nearly identical

other tasks we performed experiments on English
challenges: the output is subject to strong structural
input. Furthermore, RNN sequence-to-sequence
art results in small-data regimes [37].

024 on the Wall Street Journal (WSJ) portion of the
ces. We also trained it in a semi-supervised setting,
ser corpora from with approximately 17M sentences
e WSJ only setting and a vocabulary of 32K tokens

nts to select the dropout, both attention and residual
e Section 22 development set, all other parameters
an base translation model. During inference, we

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Table 4: The Transformer generalizes well to Englis
of WSJ)

Parser	
Vinyals & Kaiser el al. (2014) [37]	W
Petrov et al. (2006) [29]	W
Zhu et al. (2013) [40]	W
Dyer et al. (2016) [8]	W
Transformer (4 layers)	W
Zhu et al. (2013) [40]	
Huang & Harper (2009) [14]	
McClosky et al. (2006) [26]	
Vinyals & Kaiser el al. (2014) [37]	
Transformer (4 layers)	
Luong et al. (2015) [23]	
Dyer et al. (2016) [8]	

increased the maximum output length to input lengt
for both WSJ only and the semi-supervised setting.

Our results in Table 4 show that despite the lack
prisingly well, yielding better results than all previ
Recurrent Neural Network Grammar [8].

In contrast to RNN sequence-to-sequence models [
Parser [29] even when training only on the WSJ tra

7 Conclusion

In this work, we presented the Transformer, the firs
attention, replacing the recurrent layers most comm
multi-headed self-attention.

For translation tasks, the Transformer can be train
on recurrent or convolutional layers. On both W
English-to-French translation tasks, we achieve a
model outperforms even all previously reported ens

We are excited about the future of attention-based
plan to extend the Transformer to problems involvin
to investigate local, restricted attention mechanis
such as images, audio and video. Making generation

The code we used to train and evaluate our m
`tensorflow/tensor2tensor`.

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10

lish constituency parsing (Results are on Section 23

Training	WSJ 23 F1
WSJ only, discriminative	88.3
WSJ only, discriminative	90.4
WSJ only, discriminative	90.4
WSJ only, discriminative	91.7
WSJ only, discriminative	91.3
semi-supervised	91.3
semi-supervised	91.3
semi-supervised	92.1
semi-supervised	92.1
semi-supervised	92.7
multi-task	93.0
generative	93.3

ngth + 300. We used a beam size of 21 and $\alpha = 0.3$
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eviously reported models with the exception of the

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Attention Visualizations

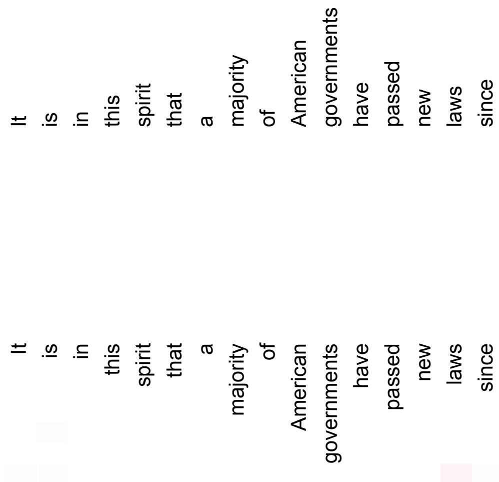

Figure 3: An example of the attention mechanis
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the verb 'making', completing the phrase 'making..
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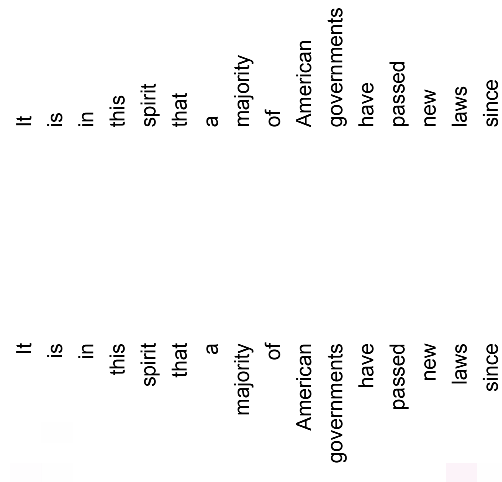
Attention Visualizations



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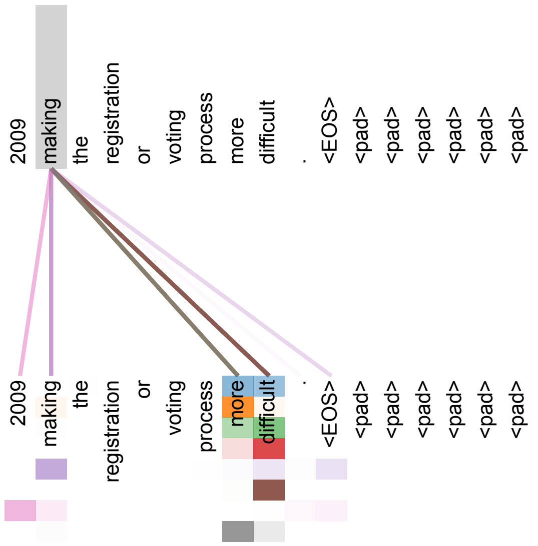

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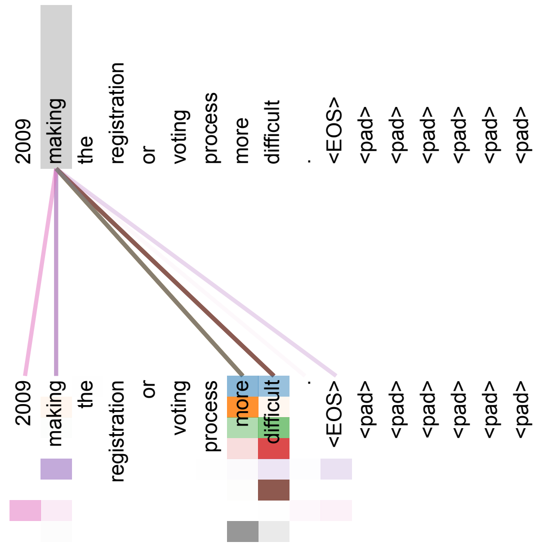

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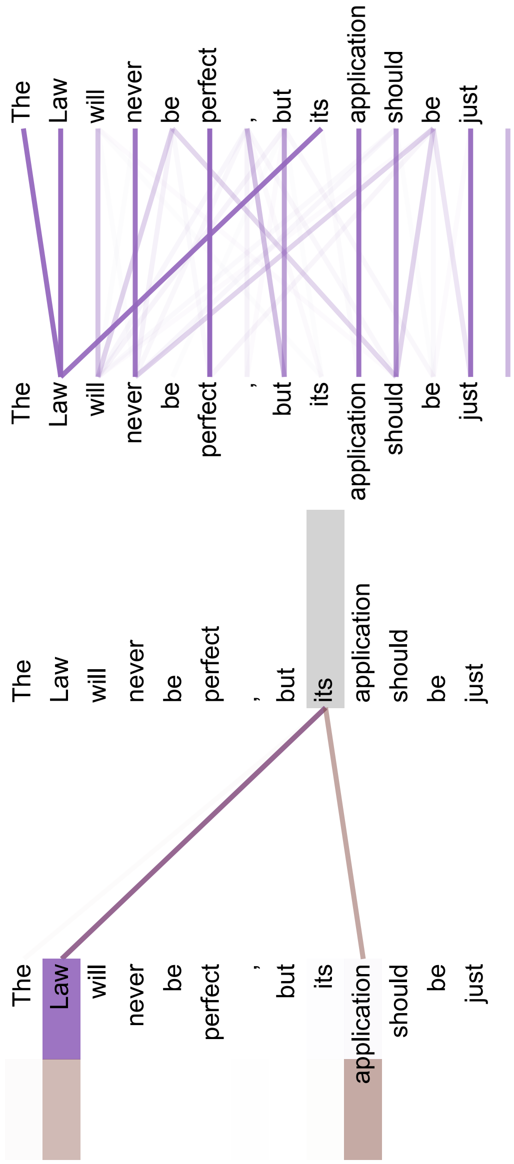

Figure 4: Two attention heads, also in layer 5 of 6,
Full attentions for head 5. Bottom: Isolated attenti
and 6. Note that the attentions are very sharp for thi

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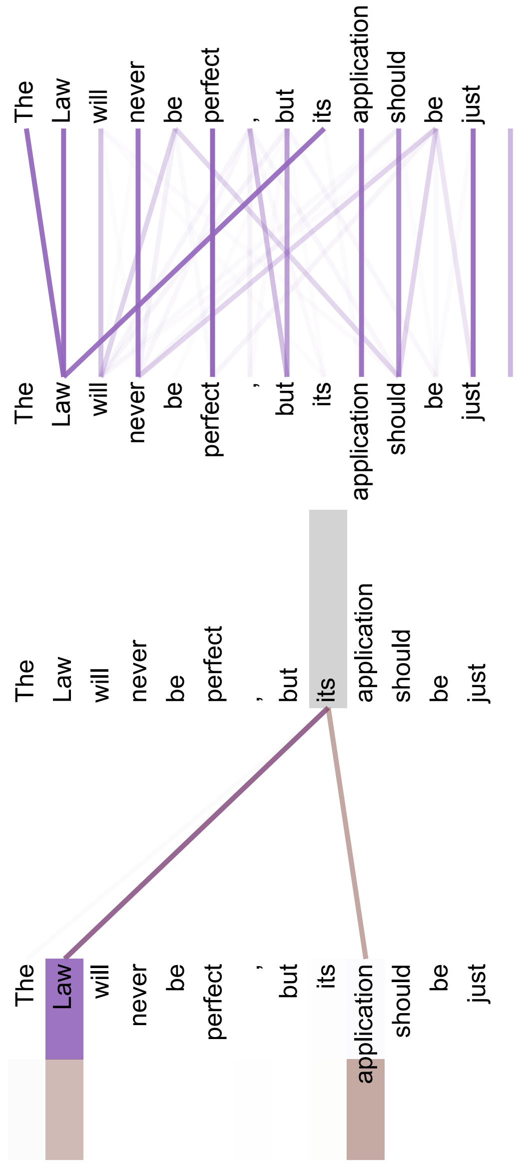

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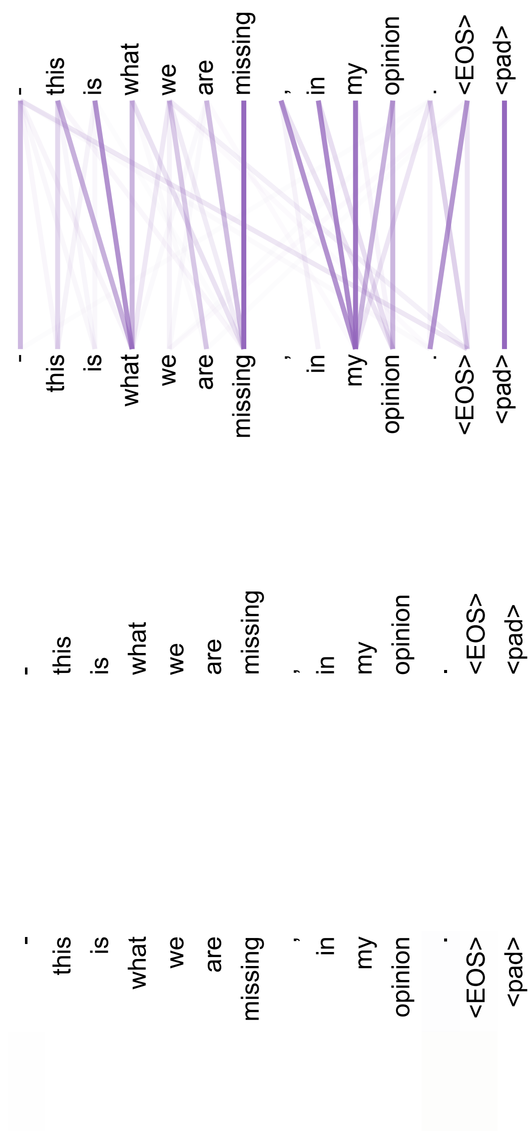

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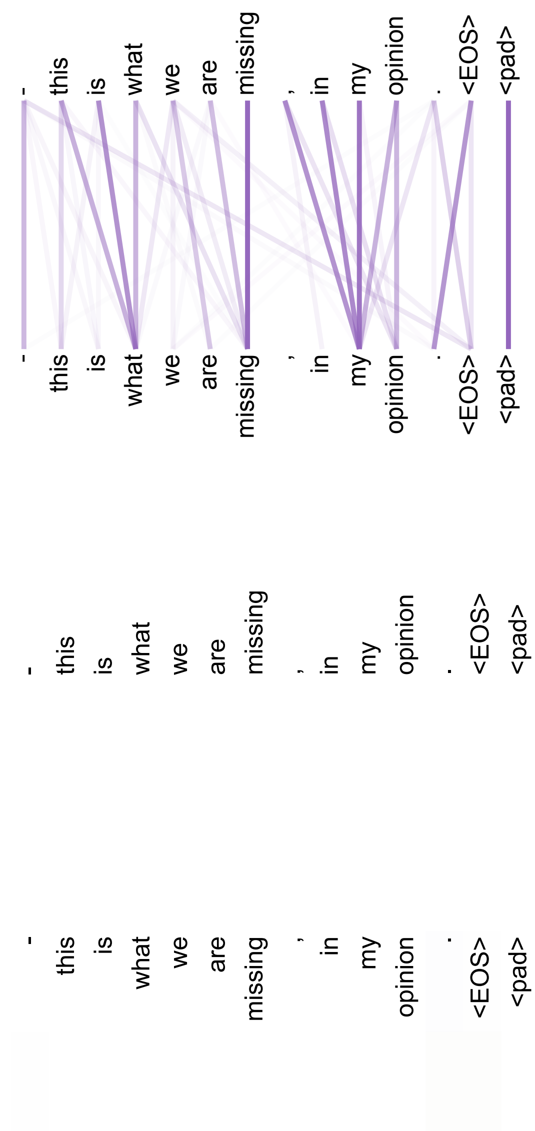
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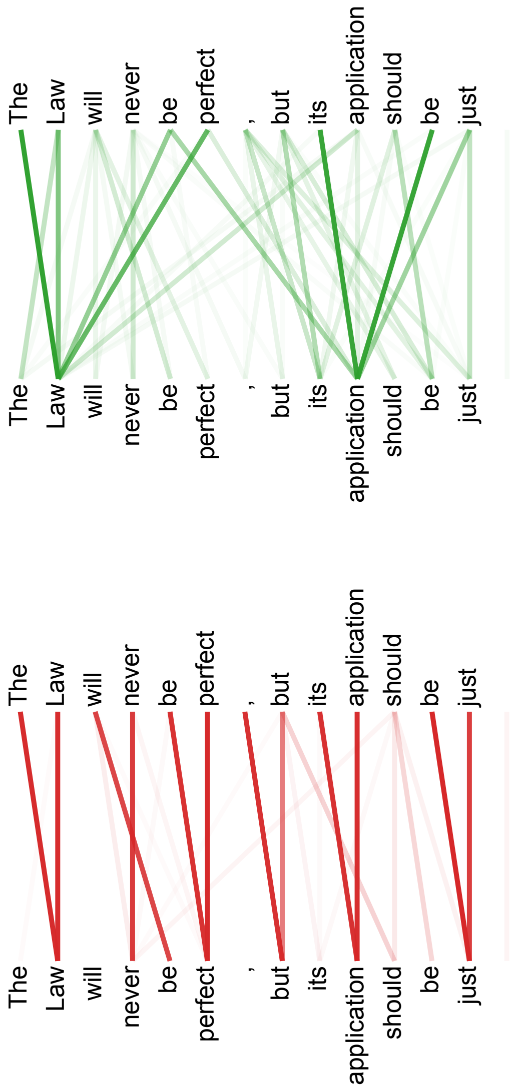

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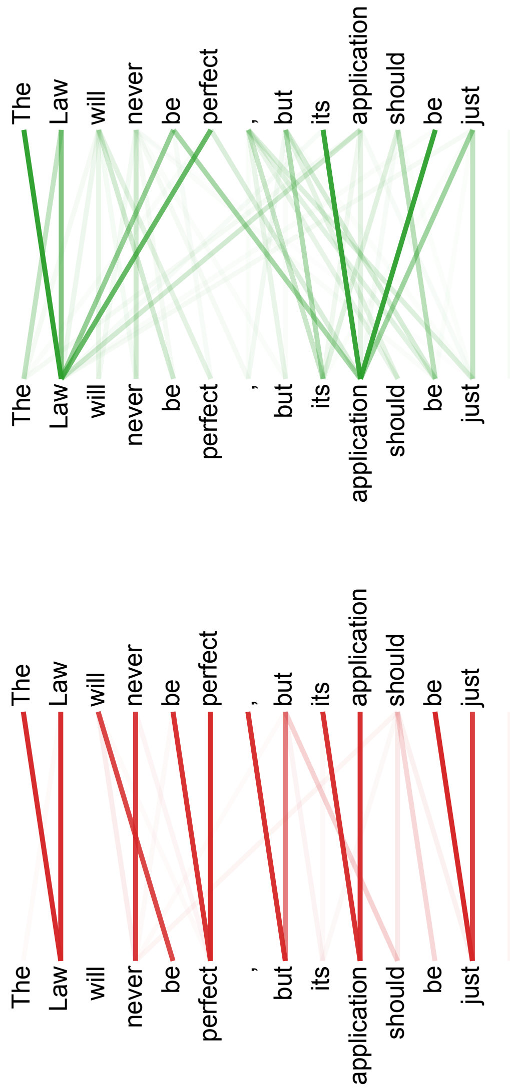

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